Introducing Google Cloud

Module 1 · Cloud basics, Google’s global infrastructure, security, sustainability and cost control



ROADMAP

What we'll cover today

01

Cloud computing defined

The five traits and how we got here

02

Service models

IaaS, PaaS, SaaS and how each is billed

03

Global infrastructure

Network, regions, zones and multi-regions

04

Sustainability

Energy use and Google's carbon-free goals

05

Security

Six layers of defense in depth

06

Openness

Open source and avoiding lock-in

07

Pricing and billing

Discounts, budgets, alerts and quotas

08

Knowledge check

Eight questions to test yourself

Five traits that make it “cloud”

The US National Institute of Standards and Technology (NIST) uses these characteristics to define cloud computing. A service needs all five, not just one or two.



On-demand self-service

Provision resources yourself through a console or API, with no ticket and no waiting.



Broad network access

Reach those resources over the internet, from any device and any location.



Resource pooling

The provider serves many customers from one large shared pool of hardware.



Rapid elasticity

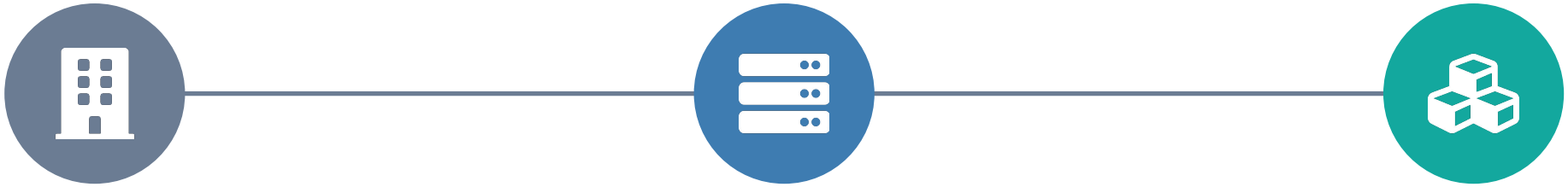
Capacity grows for peaks and shrinks again when demand falls.



Measured service

Usage is metered, so you pay only for what you consume or reserve.

Three waves that led to today's cloud



WAVE 1

Colocation

Rent floor space, power and cooling in a shared facility. You still buy, rack and run every server yourself.

WAVE 2

Virtualized data centers

Virtual machines replace physical boxes, but teams still patch, configure and scale them by hand.

WAVE 3

Containers and managed services

Automated, elastic infrastructure where the platform runs operations for you — the approach Google built for itself.

The big shift: from owning and maintaining infrastructure to consuming it as a managed, on-demand service.

Who manages what? IaaS, PaaS and SaaS

Layer	On-premises	IaaS	PaaS	SaaS
Applications	You	You	You	**Provider**
Data	You	You	You	**Provider**
Runtime and middleware	You	You	**Provider**	**Provider**
Operating system	You	You	**Provider**	**Provider**
Virtualization	You	**Provider**	**Provider**	**Provider**
Servers and storage	You	**Provider**	**Provider**	**Provider**
Networking	You	**Provider**	**Provider**	**Provider**

On Google Cloud

IaaS

Compute Engine virtual machines

PaaS

App Engine, Cloud Run — your code binds to managed libraries and runtimes

SaaS

Gmail, Google Docs, Google Workspace

Two ways to pay: allocate vs. consume



IAAS

Pay for what you allocate

You reserve capacity up front and are billed for it whether it’s busy or idle.

Example: a 4-vCPU VM that runs all month is billed for the full month, even at 5% CPU.



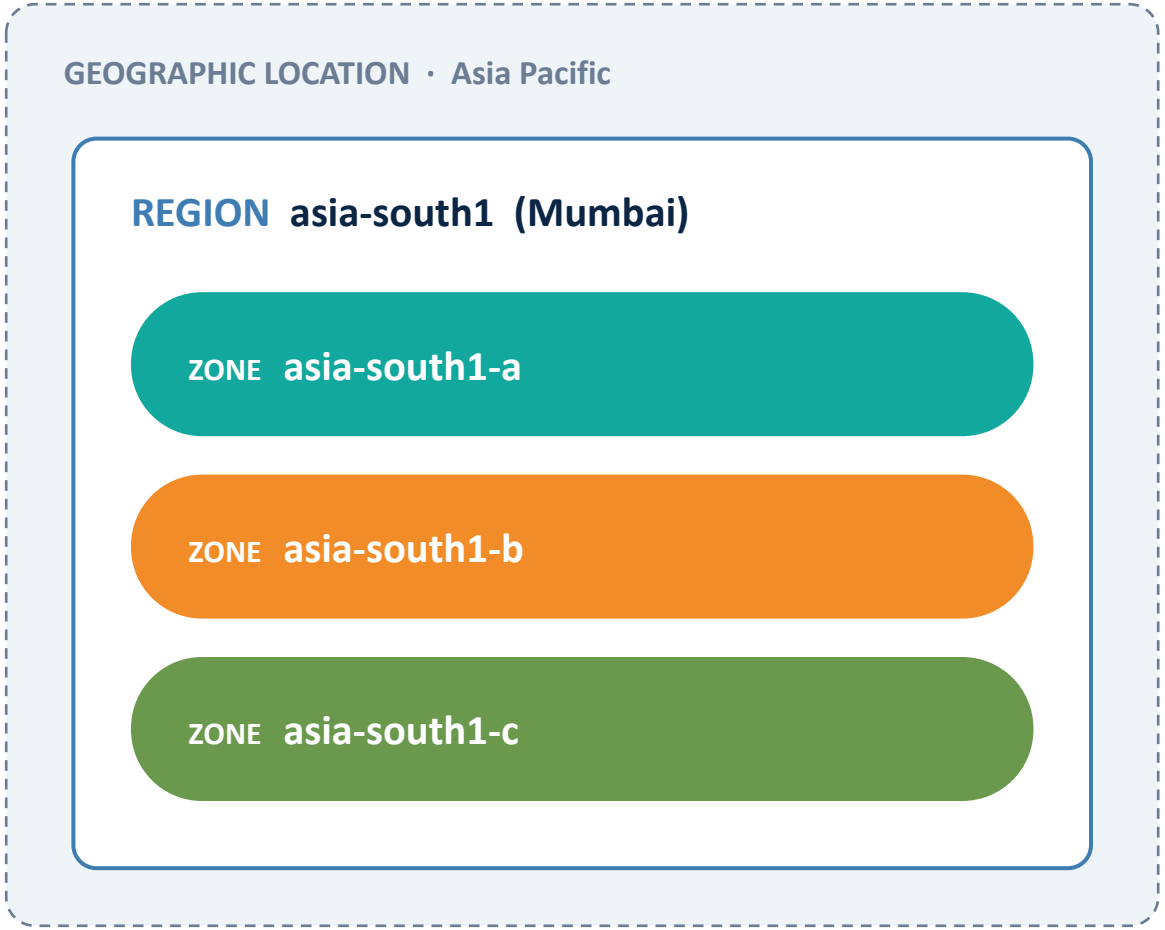
PAAS / SERVERLESS

Pay for what you use

Billing follows actual consumption — requests, compute time or data processed.

Example: a Cloud Run service with no traffic overnight costs next to nothing for those hours.

Locations, regions and zones



Zone

An isolated deployment area inside a region and a separate failure domain. Zonal resources such as VMs live here.

Region

An independent geographic area, usually made up of three or more zones. Pick one for low latency and data residency.

Multi-region

Some services, such as Cloud Storage and Spanner, can replicate data across several regions automatically.

Choosing where to run: a resilience trade-off



Single zone

Benefit

Simplest and cheapest setup.

Trade-off

Exposed if that zone has an outage.



Multiple zones

Benefit

Survives the loss of one zone with low latency between copies.

Trade-off

Still vulnerable to a region-wide event.



Multiple regions

Benefit

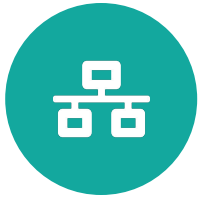
Serves users closer to home and survives a disaster that takes out a whole region.

Trade-off

More cost and design effort, data residency to consider.

40+ regions and **120+** zones worldwide, linked by Google’s private network. Current list: cloud.google.com/about/locations

A private network built for speed



Private backbone

Google has invested billions of dollars in its own global fiber and subsea cables, so traffic isn’t left to the public internet.



Close to users

Edge locations around the world let traffic enter Google’s network near the user and stay on it.



High throughput, low latency

The same network that carries Search and YouTube carries customers’ workloads.

Why it matters

Faster apps

Fewer network hops between users and services means quicker responses.

More reliable

Google controls the path end to end, from edge to data center.

Running the cloud on cleaner energy

1–2%

of the world's electricity is used by data centers, so efficiency at cloud scale really matters.



2007

Carbon neutral

Google's operations reached carbon neutrality in its first decade.

2017

100% renewable match

Annual electricity use matched with renewable purchases.

2030 goal

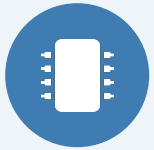
24/7 carbon-free energy

Every hour, on every grid where Google operates.

First data centers to earn ISO 14001 environmental certification

Hamina, Finland: cooled with seawater from the Gulf of Finland

Six layers of defense in depth



Hardware

Custom-designed servers, chips and network gear, a verified boot process and layered physical security.



Service deployment

Services talk over RPCs, and that traffic is encrypted as it moves between data centers.



User identity

Sign-ins are challenged based on risk, with support for second factors such as security keys.



Storage

Data is encrypted at rest with centrally managed keys, plus hardware encryption on drives.



Internet communication

The Google Front End enforces TLS, and multi-layer protection absorbs denial-of-service attacks.



Operations

Intrusion detection, limits on insider access, security keys for staff and two-person code review.

Built to keep your options open

Vendor lock-in is a common worry: once your data and apps are tied to one provider, leaving is costly. Google Cloud leans on open technologies so workloads can move.



Kubernetes and GKE

Run containerized microservices the same way on Google Cloud, other clouds or on-premises.



TensorFlow

An open-source machine learning library at the center of a large community ecosystem.



Google Cloud Observability

Monitor and log workloads that span several cloud providers from one place.

Ways to pay less for compute



Per-second billing

Compute Engine, GKE and other compute products charge by the second, not by the hour.



Sustained-use discounts

Applied automatically when a VM runs for a large share of the month.



Committed-use discounts

Commit to 1 or 3 years of usage in exchange for a lower rate.



Custom machine types

Choose exactly the vCPU and memory you need instead of a preset size.



Estimate before you build

The Google Cloud Pricing Calculator lets you model a setup and see the monthly cost before deploying anything.

cloud.google.com/products/calculator

Tools that keep spending in check

Budgets

Set a spending target for a billing account or project.

Alerts

Get notified as spend crosses thresholds such as 50%, 90% or 100%.

Reports

See where money goes by project, service and time period.

Quotas

Hard limits that stop runaway usage from errors or attacks.

Quotas are set per project and come in two kinds

Rate quota

Limits how often something can happen and resets after a time window — for example, API requests per minute.

Allocation quota

Limits how many resources a project can hold at once — for example, the number of VMs or networks.

Key takeaways

1. **Cloud = five traits** Self-service, network access, pooling, elasticity and metered billing.
2. **Pick the right model** IaaS gives control, PaaS speeds delivery, SaaS hands over the whole app.
3. **Design for failure** Spread across zones for uptime and across regions for reach and disaster recovery.
4. **Secure, green, open** Layered security, a 24/7 carbon-free goal and open-source tools.
5. **Control costs** Per-second billing, discounts, budgets, alerts and quotas.

Knowledge check

8 scenario questions on this module. Each question is followed by an answer slide, with a full answer key at the end.



QUESTION 1 OF 8

A developer launches three virtual machines at 2 a.m. using the console, without contacting anyone at the provider. Which cloud trait is this?

A Resource pooling

B On-demand self-service

C Measured service

D Broad network access

ANSWER 1 OF 8

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A Resource pooling

B **On-demand self-service** ✓

C Measured service

D Broad network access

Correct answer: B Provisioning happens instantly and without human involvement from the provider.

QUESTION 2 OF 8

An online store’s traffic triples during a festival sale, and capacity grows to match, then shrinks afterwards. Which trait does this show?

A Broad network access

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C Rapid elasticity

D On-demand self-service

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An online store’s traffic triples during a festival sale, and capacity grows to match, then shrinks afterwards. Which trait does this show?

A Broad network access

B Resource pooling

C **Rapid elasticity** ✓

D On-demand self-service

Correct answer: C Elasticity means resources expand and contract with demand.

QUESTION 3 OF 8

Your team manages the operating system, runtime and application, while the provider manages servers, storage and networking. Which model is this?

A Infrastructure as a Service

B Platform as a Service

C Software as a Service

D Colocation

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Correct answer: A IaaS hands you raw infrastructure; everything from the OS upward is yours.

QUESTION 4 OF 8

Your team uses Gmail and Google Docs and never thinks about servers, patches or code. Which service model is this?

A IaaS

B PaaS

C SaaS

D Virtualized data center

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Your team uses Gmail and Google Docs and never thinks about servers, patches or code. Which service model is this?

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C **SaaS** ✓

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Correct answer: C SaaS delivers the complete application; the customer just uses it.

QUESTION 5 OF 8

You run copies of your app in asia-south1-a and asia-south1-b. What does this mainly protect against?

A A region-wide natural disaster

B An outage in a single zone

C Denial-of-service attacks

D Unexpected billing spikes

ANSWER 5 OF 8

You run copies of your app in asia-south1-a and asia-south1-b. What does this mainly protect against?

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B. An outage in a single zone ✓

C. Denial-of-service attacks

D. Unexpected billing spikes

Correct answer: B Zones in a region are separate failure domains, so losing one zone doesn't take the app down.

QUESTION 6 OF 8

Why might a company deploy its application in regions on several continents?

A To get automatic security upgrades

B To qualify for sustained-use discounts

C To serve users with lower latency and survive a regional outage

D Because each region requires a separate account

ANSWER 6 OF 8

Why might a company deploy its application in regions on several continents?

A. To get automatic security upgrades

B. To qualify for sustained-use discounts

C. **To serve users with lower latency and survive a regional outage** ✓

D. Because each region requires a separate account

Correct answer: C Multi-region deployments bring apps closer to users and add disaster resilience.

QUESTION 7 OF 8

A project can make only a set number of API calls per minute, and the count resets afterwards. What is this?

A. Allocation quota

B. Budget alert

C. Rate quota

D. Committed-use discount

ANSWER 7 OF 8

A project can make only a set number of API calls per minute, and the count resets afterwards. What is this?

A Allocation quota

B Budget alert

C **Rate quota** ✓

D Committed-use discount

Correct answer: C Rate quotas cap how often something happens and reset after a time window.

QUESTION 8 OF 8

Which approach helps Google Cloud customers avoid vendor lock-in?

A Proprietary-only APIs

B Supporting open-source tools such as Kubernetes and TensorFlow

C Storing all data in a single zone

D Requiring 3-year contracts

ANSWER 8 OF 8

Which approach helps Google Cloud customers avoid vendor lock-in?

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B **Supporting open-source tools such as Kubernetes and TensorFlow** ✓

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Correct answer: B Open-source technologies let workloads run on other clouds or on-premises.

Up next: hands-on lab



Explore the Google Cloud Console and Cloud Shell

1. Get access to Google Cloud.
2. Use the Cloud console to create a Cloud Storage bucket.
3. Use Cloud Shell to create a Cloud Storage bucket.
4. Become familiar with Cloud Shell features.

ANSWER KEY

How did you do?

B

Q1 · On-demand self-service

Provisioning happens instantly and without human involvement from the provider.

C

Q2 · Rapid elasticity

Elasticity means resources expand and contract with demand.

A

Q3 · Infrastructure as a Service

IaaS hands you raw infrastructure; everything from the OS upward is yours.

C

Q4 · SaaS

SaaS delivers the complete application; the customer just uses it.

B

Q5 · An outage in a single zone

Zones in a region are separate failure domains, so losing one zone doesn't take the app down.

C

Q6 · To serve users with lower latency and survive a regional outage

Multi-region deployments bring apps closer to users and add disaster resilience.

C

Q7 · Rate quota

Rate quotas cap how often something happens and reset after a time window.

B

Q8 · Supporting open-source tools such as Kubernetes and TensorFlow

Open-source technologies let workloads run on other clouds or on-premises.